FUNCTION SPACES

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Fifth Semester

Contents

Chapter 1

FUNCTION SPACES

July 20th.

Quizzes 1 and 2 will account for 10% and 15% of the final grade respectively. The mid-term will account for 25% of the final grade, and the final exam will account for 50%.

1.1 An Introduction

Suppose $\mathbb{K}$ is a field, and X is a topological space. Then the set of all $\mathbb{K}$-valued functions on X, denoted by $\mathbb{K}^X$, is a vector space over $\mathbb{K}$. If we let $X = \{1, 2, \ldots, n\}$, then $\mathbb{K}^X \cong \mathbb{K}^n$, the n-dimensional vector space over $\mathbb{K}$. In this course, we will be primarily concerned with the case when $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Consider the map

$$D : \mathbb{K}^{\{1,2,\ldots,n\}} \to \mathbb{K}^{\{1,2,\ldots,n-1\}}, \quad (Df)(k) = f(k+1) - f(k) \text{ for } 1 \leq k \leq n-1, \ (Df)(n) = f(1) - f(n). \tag{1.1}$$

Here, D acts as a discrete analogue of the derivative operator by taking finite differences. We can split this finite difference map as the difference of the left shift operator and the identity operator as $D = S - I$. Since S is the left shift operator, we have $S^n = I$ so the eigenvalues of S are a subset of the n-th roots of unity. Let λ be such an eigenvalue. Then a corresponding eigenvector can be given as $\begin{pmatrix} 1 \\ \lambda \\ \vdots \\ \lambda^{n-1} \end{pmatrix}$. If $\lambda = e^{2\pi i/n}$, then we can denote the corresponding eigenvector of λ^k as χ_k. In fact, in the basis $\{\chi_1, \ldots, \chi_{n-1}, \chi_n\}$, D is a diagonal matrix. Such a basis is termed a *discrete Fourier basis*, and the coordinate vector of D with respect to this basis is called the *discrete Fourier transform* of D. The Fourier transform diagonalizes the finite differential operator. We can view χ_k as a function $\chi_k : \{1, 2, \ldots, n\} \to \mathbb{C}$ defined by $\chi_k(j) = e^{2\pi i k(j-1)/n}$ for $1 \leq j \leq n$.

Recall the Hermitian inner product on $\mathbb{C}^n$ given by $\langle x, y \rangle = \sum_{i=1}^n x_i \overline{y_i}$. Viewing $\mathbb{C}^n$ as the function space $\mathbb{C}^{\{1,2,\ldots,n\}}$, we can endow it with the inner product $\langle f, g \rangle = \sum_{k=1}^n f(k)\overline{g(k)}$, which gives the 2-norm $\|f\|_2 = \sqrt{\sum_{i=1}^n |f(i)|^2}$. The discrete Fourier basis is an orthogonal basis with respect to this inner product. Normalizing via division by $\sqrt{n}$ makes it an orthonormal basis. Thus, if we have $f \in \mathbb{C}^{\{1,2,\ldots,n\}}$ which we wish to write as $f = \sum_{j=1}^n a_j \chi_j$, then we can compute the coefficients a_j as

$$a_j = \frac{1}{n}\langle f, \chi_j \rangle = \frac{1}{n}\sum_{r=1}^n f(r) e^{-2\pi i j(r-1)/n}. \tag{1.2}$$

This is the discrete Fourier transform, as before. If we look at the norm, then

$$\|f\|_2^2 = \left\| \sum_{j=1}^n a_j \chi_j \right\|_2^2 = n \sum_{j=1}^n |a_j|^2 \implies \frac{1}{n}\|f\|_2^2 = \sum_{j=1}^n |a_j|^2. \tag{1.3}$$

This is known as *Plancherel's theorem.*

All of the above can be generalized to the continous case; let $L^2[0,1]$ be the space of square integrable functions on $[0,1]$. If $(a_n)_{n\in\mathbb{Z}}$ is the Fourier series of f, then $\int_0^1 |f(x)|^2\,dx = \sum_{n\in\mathbb{Z}} |a_n|^2$, which is the continuous analogue of Plancherel's theorem.

1.2 Metric and Normed Spaces

July 23rd.

The following theorem relates the notion of compactness, that every cover has a finite subcover, to many other properties of metric spaces.

Theorem 1.1. *Let (X,d) be a metric space. The following are equivalent:*

1. *X is compact.*
2. *Every nested sequence of nonempty closed sets $F_1 \supseteq F_2 \supseteq \cdots$ has nonempty intersection, that is,*
$$\bigcap_{n=1}^{\infty} F_n \neq \emptyset. \tag{1.4}$$
3. *Every countable family of closed sets with the finite intersection property has nonempty intersection.*
4. *Every sequence in X has a convergent subsequence.*
5. *X is complete and totally bounded.*

Proof. For $1 \implies 2$, let $\{F_n\}_{n=1}^{\infty}$ be a nested sequence of nonempty closed sets with $F_1 \supseteq F_2 \supseteq \cdots$. If $\bigcap_{n=1}^{\infty} F_n = \emptyset$, then $\{X \setminus F_n\}_{n=1}^{\infty}$ is an open cover of X. By compactness, there exists a finite subcover. Since the F_n are nested,
$$X = \bigcup_{i=1}^{N} (X \setminus F_i) = X \setminus F_N. \tag{1.5}$$
Hence $F_N = \emptyset$, a contradiction.

For $2 \implies 3$, let $\{F_n\}_{n=1}^{\infty}$ be a countable family of closed sets with the finite intersection property. Define
$$G_n = \bigcap_{i=1}^{n} F_i. \tag{1.6}$$
Then $G_1 \supseteq G_2 \supseteq \cdots$ is a nested sequence of nonempty closed sets. By 2, $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$. Since
$$\bigcap_{n=1}^{\infty} G_n = \bigcap_{n=1}^{\infty} F_n, \tag{1.7}$$
the result follows.

For $3 \implies 4$, let S_0 be the set of elements in $(x_i)_{i=1}^{\infty}$, a sequence in X. Similarly, let S_1 be the set of elements in $(x_i)_{i=2}^{\infty}$, and define S_n similarly. Then
$$\overline{S}_0 \supseteq \overline{S}_1 \supseteq \cdots \supseteq \overline{S}_n \supseteq \cdots. \tag{1.8}$$
By 3, $\bigcap_{n=1}^{\infty} \overline{S}_n \neq \emptyset$, so there exists $x' \in \bigcap_{n=1}^{\infty} \overline{S}_n$. Now, $B(x',1) \cap S_1$ cannot be empty, so pick x_{n_1} from this intersection. Again, $B(x',1/2) \cap S_{n_1}$ cannot be empty, so pick x_{n_2} from this intersection. Continue inductively to obtain the x_{n_k}'s. One can verify that $x_{n_k} \to x'$.

For $4 \implies 5$, every Cauchy sequence has a convergent subsequence, and hence converges. Thus, X is complete. Suppose X is not totally bounded. Then there exists $\alpha > 0$ such that X has no finite covering by balls of radius α. Pick $x_1 \in X$. Since $B(x_1,\alpha) \neq X$, pick $x_2 \in X \setminus B(x_1,\alpha)$. Again,
$$B(x_1,\alpha) \cup B(x_2,\alpha) \neq X. \tag{1.9}$$

Similarly, pick x_3 outside the previous balls, and continue inductively. Since $d(x_m, x_n) \geq \alpha$ for $m \neq n$, there is no convergent subsequence, contradicting 4. Hence X is totally bounded.

For 5 $\implies$ 1, we prove the contrapositive. Assume X is not compact. If it is not totally bounded, we are done; further assume that it is totally bounded. Then there exists an open cover $\{U_\alpha\}_{\alpha\in\Lambda}$ with no finite subcover. Let $\varepsilon = 1/2$ and pick a finite ε-net. Pick x_1 such that $B(x_1, 1/2)$ is not covered by finitely many U_α. Since $B(x_1, 1/2)$ is totally bounded, pick $x_2 \in B(x_1, 1/2)$ such that $B(x_2, 1/4)$ cannot be covered by finitely many U_α. Inductively, pick $x_n \in B(x_{n-1}, 1/2^{n-1})$ such that $B(x_n, 1/2^n)$ cannot be covered by finitely many U_α. Then

$$d(x_n, x_m) \leq \sum_{k=m}^{n-1} \frac{1}{2^k}, \tag{1.10}$$

so (x_n) is a Cauchy sequence. Since X is complete, $x_n \to x$ for some $x \in X$. Let $x \in U_\alpha$. Then there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq U_\alpha$. Moreover, $d(x_n, x) < \varepsilon/2$ for n sufficiently large. Hence

$$B(x_n, 2^{-n}) \subseteq B(x_n, \varepsilon/2) \subseteq B(x, \varepsilon) \subseteq U_\alpha, \tag{1.11}$$

for all sufficiently large n, since $2^{-n} < \varepsilon/2$. This contradicts the choice of $B(x_n, 2^{-n})$. Hence X is not complete. ■

We now look at some examples of non-trivial function spaces; consider $C[0,1]$, the space of complex-valued continuous functions on $[0,1]$. This is a vector space over $\mathbb{R}$ with pointwise addition and scalar multiplication. We can define a norm on this space as $\|f\|_\infty = \sup_{x\in[0,1]} |f(x)|$. This makes $C[0,1]$ into a normed space.

Definition 1.2. Let X be a $\mathbb{K}$-vector space. A *norm* on X is a function $\|\cdot\| : X \to [0,\infty)$ such that for all $x, y \in X$ and $\alpha \in \mathbb{K}$,

1. $\|x\| = 0$ if and only if $x = 0$ (positive definiteness).
2. $\|\alpha x\| = |\alpha| \, \|x\|$ (homogeneity).
3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

One may verify that $\|\cdot\|_\infty$ is indeed a norm on $C[0,1]$.

1.3 The Lebesgue Integral

July 27th.

Definition 1.3. A function $s : [0,1] \to \mathbb{C}$ is called a *step function* if there exists a partition $P : 0 = x_0 < x_1 < \cdots < x_n = 1$ of $[0,1]$ such that s is constant on each interval (x_{i-1}, x_i) for $1 \leq i \leq n$. That is, $s(x) = c_k$ if $x \in (x_{k-1}, x_k)$ where c_k are constants for $1 \leq k \leq n$.

We impose that the integral of such a step function is given by

$$\int_0^1 s(x)\,dx = \sum_{k=1}^{n} c_k(x_k - x_{k-1}). \tag{1.12}$$

The values of the step function on the partition points x_i do not affect the value of the integral.

Proposition 1.4. *The set of step functions on $[0,1]$ is a complex vector space under pointwise addition and scalar multiplication.*

The proof is straightforward and left as an exercise.

Proposition 1.5. *The set of step functions on* $[0,1]$ *such that* $\int_0^1 |s(x)|\,dx = 0$ *is a subspace of the vector space of step functions on* $[0,1]$.

Again, this proof is rudimentary and left as an exercise. The reader is also urged to show that $\int_0^1 s(x)\,dx$ is well-defined and independent of the choice of partition P.

Definition 1.6. A set $S \subseteq [0,1]$ is said to have *content zero* or measure zero if for every $\varepsilon > 0$, there exists countably many open intervals $(a_n, b_n)_{n=1}^{\infty}$ such that $S \subseteq \bigcup_{n=1}^{\infty}(a_n, b_n)$ and $\sum_{n=1}^{\infty}(b_n - a_n) < \varepsilon$.

As a trivial example, any finite subset of $[0,1]$ has content zero. Similarly, any countable subset of $[0,1]$ has content zero. The Cantor set is an example of an uncountable set with content zero. One may show that the countable union of sets of content zero has content zero.

Definition 1.7. A sequence of functions $f_n : [0,1] \to \mathbb{R}$ is said to be increasing if $f_n(x) \leq f_{n+1}(x)$ for all $x \in [0,1]$. Similarly, a sequence of decreasing functions is defined.

Note that a property is said to be hold *almost everywhere* if it holds for all $x \in [0,1]$ except for a negligible set (a set of content zero).

Definition 1.8. We say $f_n \to f$ pointwise almost everywhere if there exists a set $S \subseteq [0,1]$ of content zero such that $f_n(x) \to f(x)$ for all $x \in [0,1] \setminus S$. That is, f_n converges to f pointwise except on a negligible set.

Theorem 1.9 (The *proto monotone convergence theorem*)**.** *Let* $\{s_n\}_{n=1}^{\infty}$ *be a decreasing seqeucence of non-negative step functions on* $[0,1]$ *such that* $s_n \to 0$ *pointwise almost everywhere. Then* $\int_0^1 s_n(x)\,dx \to 0$ *as* $n \to \infty$.

Proof. Let $E := \{x \in [0,1] \mid s_n(x) \not\to 0\}$. Then E has content zero. Define D to be the set of endpoints of intervals in the partitions corresponding to the s_n's. Then D is countable, and hence has content zero. Let $F = D \cup E$ is negligible. Let $x \in [0,1] \setminus F$. Then there exists $N(x) \in \mathbb{N}$ such that $s_n(x) = 0$ for all $n \geq N(x)$. Since $x \notin D$, there exists an open interval $B(x)$ such that $s_N(t) < \varepsilon$ for all $t \in B(x)$. As $\{s_n\}$ is decreasing, we have $s_n(t) < \varepsilon$ for all $t \in B(x)$ and $n \geq N(x)$. For every $x \in [0,1] \setminus F$, we have a $B(x)$. Find a sequence of open intervals such that $F \subseteq \bigcup_{n=1}^{\infty} F_n$ and $\sum_{n=1}^{\infty} |F_n| < \varepsilon$. Then

$$[0,1] \subseteq \left(\bigcup_{x \in [0,1] \setminus F} B(x) \right) \cup \left(\bigcup_{n=1}^{\infty} F_n \right), \tag{1.13}$$

an open cover. By compactness, there exists a finite subcover, so

$$[0,1] \subseteq \bigcup_{i=1}^{p} B(x_i) \cup \bigcup_{r=1}^{q} F_r. \tag{1.14}$$

Call the left union P and the right union Q. Then

$$\int_Q s_n(x)\,dx := \int_0^1 \chi_Q(x) s_n(x)\,dx \leq M \int \sum_{i=1}^{q} |F_i|\,dx < M\varepsilon. \tag{1.15}$$

Similarly,

$$\int_P s_n(x)\,dx := \int_0^1 \chi_P(x) s_n(x)\,dx \leq \varepsilon \quad \text{for all } n \geq \max\{N(x_i) \mid 1 \leq i \leq p\}. \tag{1.16}$$

Combining gives

$$\int_0^1 s_n(x)\,dx = \int_P s_n(x)\,dx + \int_Q s_n(x)\,dx < (M+1)\varepsilon. \tag{1.17}$$

Thus, $\int_0^1 s_n(x)\,dx \to 0$ as $n \to \infty$. ■

From here, we can immediately note that $\rho : \mathcal{S}[0,1] \to \mathbb{C}$, map from the set of step functions, defined by $s \mapsto \int_0^1 s(x)\,dx$ is a linear functional. We can also define a semi-norm on the set of step functions as $\|s\|_1 = \int_0^1 |s(x)|\,dx$. This is a semi-norm since $\|s\|_1 = 0$ does not imply that $s = 0$; it only implies that $s = 0$ almost everywhere. We can define an equivalence relation on the set of step functions by $s \sim t$ if and only if $s = t$ almost everywhere. Then we can define a norm on the quotient space $\mathcal{S}[0,1]/\sim$ as $\|s\|_1 = \int_0^1 |s(x)|\,dx$. This is indeed a norm since $\|s\|_1 = 0$ implies that $s = 0$ in the quotient space.

The normed space $(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$ is not complete; there is a much larger class of functions that are (Lebesgue) integrable. Henceforth, we define $L^1[0,1]$ to be the completion of $(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$. The elements of $L^1[0,1]$ are called *Lebesgue integrable functions*.

July 30th.

Theorem 1.10. *Let $\{t_n\}_{n=1}^\infty$ be a sequence of step functions on $[0,1]$ such that*

1. *There is a function f such that $t_n \uparrow f$ pointwise almost everywhere.*
2. *The sequence $\{\int_0^1 t_n(x)\,dx\}_{n=1}^\infty$ converges.*

Then, for any step function t such that $t(x) \le f(x)$ almost everywhere, we have $\int_0^1 t(x)\,dx \le \lim_{n\to\infty} \int_0^1 t_n(x)\,dx$.

Proof. Define s_n as

$$s_n(x) = \begin{cases} t(x) - t_n(x), & \text{if } t(x) \ge t_n(x), \\ 0, & \text{if otherwise.} \end{cases} \tag{1.18}$$

That is, $s_n(x) = \max\{t(x) - t_n(x), 0\}$ for all $n \ge 1$. Also, $s_n(x) \ge t(x) - t_n(x)$ almost everywhere, so

$$\int_0^1 s_n(x)\,dx \ge \int_0^1 t(x)\,dx - \int_0^1 t_n(x)\,dx. \tag{1.19}$$

Taking limits on both sides gives

$$\lim_{n\to\infty} \int_0^1 t_n(x)\,dx \ge \int_0^1 t(x)\,dx \tag{1.20}$$

since $\lim_{n\to\infty} \int_0^1 s_n(x)\,dx = 0$ by the proto monotone convergence theorem. ∎

1.3.1 Upper Functions and L^1

Recall the linear functional $\rho : \mathcal{S}[0,1] \to \mathbb{R}$ (or $\mathbb{C}$) defined by $s \mapsto \int_0^1 s(x)\,dx$. We can extend this functional to a larger class of functions.

Definition 1.11. A function $f : [0,1] \to \mathbb{R}$ is said to be an upper function on $[0,1]$ if there exists an increasing sequence of step functions $\{s_n\}_{n=1}^\infty$ such that $s_n \uparrow f$ pointwise almost everywhere, and $\int_0^1 s_n(x)\,dx$ converges.

At this point, we can define the integral of an upper function f as $\int_0^1 f dx := \lim_{n\to\infty} \int_0^1 s_n(x)\,dx$. The right hand side is well-defined by the previous theorem, but we need to show that it is independent of the choice of the sequence $\{s_n\}_{n=1}^\infty$ that converges to f pointwise almost everywhere. This is the content of the following theorem.

Theorem 1.12. *$\int_0^1 f dx$ is well-defined for an upper function f.*

Proof. Let $\{s_n\}_{n=1}^\infty$ and $\{t_n\}_{n=1}^\infty$ be two increasing sequences of step functions such that $s_n \uparrow f$ and $t_n \uparrow f$ pointwise almost everywhere. Since $s_n \le f$ almost everywhere, we have $\int_0^1 s_n(x)\,dx \le \lim_{k\to\infty} \int_0^1 t_k(x)\,dx$ by the previous theorem. Thus,

$$\lim_{n\to\infty} \int_0^1 s_n(x)\,dx \le \lim_{k\to\infty} \int_0^1 t_k(x)\,dx. \tag{1.21}$$

By a symmetry argument, we have $\lim_{n\to\infty} \int_0^1 t_n(x)\,dx = \lim_{k\to\infty} \int_0^1 s_k(x)\,dx$. Hence, $\int_0^1 f dx$ is well-defined. ∎

Let $U[0,1]$ denote the set of upper functions on $[0,1]$. Trivially, $\mathcal{S}[0,1] \subseteq U[0,1]$.

Theorem 1.13. *Let $f, g \in U[0,1]$, and $\lambda \ge 0$. Then $f+g \in U[0,1]$ and $\lambda f \in U[0,1]$. Moreover,*

$$\int_0^1 (f+g)\,dx = \int_0^1 f\,dx + \int_0^1 g dx, \quad \int_0^1 (\lambda f)\,dx = \lambda \int_0^1 f\,dx. \tag{1.22}$$

If $f \le g$ almost everywhere, then $\int_0^1 f dx \le \int_0^1 g dx$.

Note that it is clear that all these properties are true for step functions.

Proof. Suppose $s_n \uparrow f$ and $t_n \uparrow g$ almost everywhere, and that $\int_0^1 s_n(x)\,dx$ and $\int_0^1 t_n(x)\,dx$ converge. Then $s_n + t_n \uparrow f+g$ almost everywhere, and $\lambda s_n \uparrow \lambda f$ almost everywhere. By the linearity of the integral for step functions, we have

$$\int_0^1 (s_n+t_n)\,dx = \int_0^1 s_n\,dx + \int_0^1 t_n\,dx, \quad \int_0^1 (\lambda s_n)\,dx = \lambda \int_0^1 s_n\,dx. \tag{1.23}$$

Now suppose $f \le g$ almost everywhere. Then $s_n \le g$ almost everywhere, so $\int_0^1 s_n(x)\,dx \le \int_0^1 g\,dx$ by the previous theorem. Taking limits gives $\int_0^1 f\,dx \le \int_0^1 g\,dx$. ∎

Theorem 1.14. *If $f, g \in U[0,1]$ and $f = g$ almost everywhere, then $\int_0^1 f\,dx = \int_0^1 g\,dx$.*

Proof. Note that $f \le g$ almost everywhere and $g \le f$ almost everywhere. Hence, $\int_0^1 f\,dx \le \int_0^1 g\,dx$ and $\int_0^1 g\,dx \le \int_0^1 f\,dx$. Thus, $\int_0^1 f\,dx = \int_0^1 g\,dx$. ∎

Theorem 1.15. *If $f, g \in U[0,1]$, then* $\max\{f,g\}, \min\{f,g\} \in U[0,1]$.

August 3rd.

Theorem 1.16. *Let $f : [0,1] \to \mathbb{R}$ be continuous and bounded. Then $f \in U[0,1]$ and the Lebesgue integral of f is equal to the Riemann integral of f.*

Proof. Let $\mathcal{P}_n : x_0 < \cdots < x_{2^n}$ be a partition of $[0,1]$ into 2^n equal subintervals of length 2^{-n}. Let $m_k = \inf\{f(x) \mid x \in [x_{k-1}, x_k]\}$ for $1 \le k \le 2^n$, and define the step function s_n as

$$s_n(x) = \begin{cases} m_k, & \text{if } x \in (x_{k-1}, x_k], \\ m_1, & \text{if } x = x_0. \end{cases} \tag{1.24}$$

Infer that $s_{n+1} \ge s_n$ since

$$\inf\{f(x) \mid x \in [x_{k-1}, x_k]\} \le \inf\{f(x) \mid x \in [x_{2k-2}, x_{2k}]\} \le \inf\{f(x) \mid x \in [x_{2k-2}, x_{2k-1}]\}. \tag{1.25}$$

Also observe that $s_n(x) \to f(x)$ for all $x \in [0,1]$ for which f is continuous. Given $\varepsilon > 0$, there exists $\delta > 0$ such that for all $y \in (x-\delta, x+\delta) \cap [0,1]$, we have $|f(y) - f(x)| < \varepsilon$. Choose N such that the partition $\mathcal{P}_N$ has a subinterval $[x_{k-1}, x_k]$ containing x and lying in $(x-\delta, x+\delta)$. Then

$$s_N(x) = m_k \le f(x) \le \inf\{f(y) \mid y \in (x-\delta, x+\delta)\} \le m_k + \varepsilon = s_N(x) + \varepsilon. \tag{1.26}$$

That is, $|s_N(x) - f(x)| < \varepsilon$. Since ε was arbitrary, $s_n(x) \to f(x)$ for all $x \in [0,1]$. Finally, observe that $\int_0^1 s_n(x)\,dx$ is the lower Riemann sum of f with respect to the partition $\mathcal{P}_n$. Since f is Riemann integrable, we have $\int_0^1 s_n(x)\,dx \to \int_0^1 f(x)\,dx$ as $n \to \infty$. Hence, $f \in U[0,1]$ and the Lebesgue integral of f is equal to the Riemann integral of f. ■

Given general upper functions $u, v \in U[0,1]$, the function $u - v$ may not neccesarily lie in $U[0,1]$. From here, we define the following.

Definition 1.17. $L^1[0,1]$ is the set of functions $f : [0,1] \to \mathbb{R}$ such that $f = u - v$ for some $u, v \in U[0,1]$. It is termed the space of Lebesgue integrable functions on $[0,1]$. f is said to be *Lebesgue integrable* here, with the Lebesgue integral of f defined as

$$\int_0^1 f\,dx := \int_0^1 u\,dx - \int_0^1 v\,dx. \tag{1.27}$$

Theorem 1.18. *Let $u, v, w, z \in U[0,1]$ be such that $u - v = w - z$ almost everywhere. Then $\int_0^1 u\,dx - \int_0^1 v\,dx = \int_0^1 w\,dx - \int_0^1 z\,dx$.*

Proof. Note that $u + z = w + v$ almost everywhere. Hence, $\int_0^1 u\,dx + \int_0^1 z\,dx = \int_0^1 w\,dx + \int_0^1 v\,dx$. Rearranging gives the desired result. ■

Thus, the Lebesgue integral of $f \in L^1[0,1]$ is well-defined.

Theorem 1.19. *Let $f, g \in L^1[0,1]$. Then*

1. *$af + bg \in L^1[0,1]$ for all $a, b \in \mathbb{R}$, and $\int_0^1 (af + bg)\,dx = a\int_0^1 f\,dx + b\int_0^1 g\,dx$.*
2. *$\int_0^1 f\,dx \geq 0$ if $f \geq 0$ almost everywhere.*
3. *$\int_0^1 f\,dx \geq \int_0^1 g\,dx$ if $f \geq g$ almost everywhere.*
4. *$\int_0^1 f\,dx = \int_0^1 g\,dx$ if $f = g$ almost everywhere.*

Proof.

1. Let $a^+ = \max\{a, 0\}$ and $a^- = \max\{-a, 0\}$. Then $af = a(u - v) = au - av = (a^+u + a^-v) - (a^-u + a^+v)$. Similarly, $bg = b(w - z) = (b^+w + b^-z) - (b^-w + b^+z)$. Hence, $af + bg \in L^1[0,1]$. The linearity of the integral follows from the linearity of the integral on upper functions.
2. Let $f = u - v$ for some $u, v \in U[0,1]$. Then $u \geq v$ almost everywhere, so $\int_0^1 u\,dx \geq \int_0^1 v\,dx$. Hence, $\int_0^1 f\,dx = \int_0^1 u\,dx - \int_0^1 v\,dx \geq 0$.
3. Follows from 2.
4. Follows from 2 and 3.

■

If f is a real-valued function, we can define the positive and negative parts of f as $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$. Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Theorem 1.20. *If $f, g \in L^1[0,1]$, then $f^+, f^-, |f|, \max\{f, g\}, \min\{f, g\} \in L^1[0,1]$. Moreover,*

$$\left|\int_0^1 f\,dx\right| \leq \int_0^1 |f|\,dx. \tag{1.28}$$

Proof. Let $f = u - v$ for some $u, v \in U[0,1]$. Then $f^+ = \max\{u-v, 0\} = \max\{u,v\} - v \in L^1[0,1]$ and $f^- = \max\{-f, 0\} = \max\{v-u, 0\} = \max\{u,v\} - u \in L^1[0,1]$. Hence, $|f| = f^+ + f^- \in L^1[0,1]$. Similarly, $\max\{f,g\}, \min\{f,g\} \in L^1[0,1]$. Finally,

$$-\int_0^1 |f|\, dx \leq -\int_0^1 f^-\, dx = \int_0^1 f^+ - f^-\, dx = \int_0^1 f\, dx \leq \int_0^1 f^+\, dx \leq \int_0^1 |f|\, dx. \tag{1.29}$$

■

August 6th.
We also recall the following theorem from real analysis.

Theorem 1.21 (*Lebesgue-Vitali theorem*)**.** *$f : [0,1] \to \mathbb{R}$ is Riemann integrable if and only if f is continuous almost everywhere, and bounded.*

Theorem 1.16 is a generalization of this, with the condition of f being continuous replaced with f being continuous almost everywhere. Of course, this theory extends to complex-valued functions.

Definition 1.22. A function $f : [0,1] \to \mathbb{C}$ is said to be Lebesgue integrable, denoted $f \in L^1([0,1];\mathbb{C}) (\equiv L^1[0,1])$ if $\Re(f), \Im(f) \in L^1([0,1];\mathbb{R})$. The Lebesgue integral of f is defined as

$$\int_0^1 f\, dx := \int_0^1 \Re(f)\, dx + i \int_0^1 \Im(f)\, dx \tag{1.30}$$

where $\Re(f) = \frac{f + \overline{f}}{2}$ and $\Im(f) = \frac{f - \overline{f}}{2i}$.

Theorem 1.23. *If $f \in L^1([0,1];\mathbb{C})$, then $|f| \in L^1([0,1];\mathbb{R})$ and $\left|\int_0^1 f\, dx\right| \leq \int_0^1 |f|\, dx$.*

Proof. Note that $h_1 = \Re(f), h_2 = \Im(f) \in L^1([0,1];\mathbb{R})$, so $|\Re(f)|, |\Im(f)| \in L^1([0,1];\mathbb{R})$. Let s_n and t_n be a sequence of step functions that converge to h_1 and h_2 from below. Then,

$$\sqrt{s_n^2 + t_n^2} \leq |s_n| + |t_n| \implies \int_0^1 |f(x)|\, dx := \lim_{n\to\infty} \int_0^1 \sqrt{s_n^2 + t_n^2}\, dx \leq \int_0^1 |h_1(x)|\, dx + \int_0^1 |h_2(x)|\, dx < \infty. \tag{1.31}$$

Thus, $|f| \in L^1([0,1];\mathbb{R})$. For the inequality, we can break down f as $f(x) = u(x)\,|f(x)|$, where $|u(x)| = 1$ for all $x \in [0,1]$. Thus, $\overline{u(x)} f(x) = |f(x)|$ ■

1.3.2 Convergence and Density

Recall the equivalence relation $\sim$ used to define the quotient space $\mathcal{S}[0,1]/\sim$, which was then extended to form $L^1[0,1]$. We claim that $\mathcal{S}[0,1]/\sim \subseteq L^1[0,1]$ is dense in $(L^1[0,1], \|\cdot\|_1)$.

Lemma 1.24. *$(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$ is dense in $(C[0,1], \|\cdot\|_1)$.*

Proof. Let $f \in C[0,1]$. Then f is bounded, so there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in [0,1]$. Let $\varepsilon > 0$. By the uniform continuity of f, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. Choose $N \in \mathbb{N}$ such that $2^{-N} < \delta$. Consider the partition $\mathcal{P}_N : x_0 < x_1 < \cdots < x_{2^N}$ of $[0,1]$ into 2^N equal subintervals of length 2^{-N}. Define the step function s_N as

$$s_N(x) = f(x_{k-1}) \quad \text{for } x \in (x_{k-1}, x_k], \quad 1 \leq k \leq 2^N. \tag{1.32}$$

Then,

$$\|f - s_N\|_1 = \int_0^1 |f(x) - s_N(x)|\, dx = \sum_{k=1}^{2^N} \int_{x_{k-1}}^{x_k} |f(x) - f(x_{k-1})|\, dx < 2^N 2^{-N} \varepsilon = \varepsilon. \tag{1.33}$$

Hence, $(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$ is dense in $(C[0,1], \|\cdot\|_1)$. ■

Lemma 1.25. *$(C[0,1], \|\cdot\|_1)$ is dense in $(L^1[0,1], \|\cdot\|_1)$.*

Theorem 1.26. *Let $\{s_n\}_{n\geq 1}$ be an increasing sequence of step functions such that $\lim_{n\to\infty}\int_0^1 s_n(x)\,dx$ exists. Then s_n converges pointwise almost everywhere to some upper function $f \in U[0,1]$ and $\int_0^1 f\,dx = \lim_{n\to\infty}\int_0^1 s_n(x)\,dx = \int_0^1 \lim_{n\to\infty} s_n(x)\,dx$.*

Proof. Without the loss of generality, we can assume that $s_n \geq 0$ for all $n \geq 1$; also Let $M = \lim_{n\to\infty}\int_0^1 s_n(x)\,dx$. Define $D = \{x \in [0,1] \mid s_n(x) \uparrow \infty \text{ or diverges}\}$. We wish to show that D has content zero. Let $\varepsilon > 0$. Further define $t_n(x) = \left\lfloor \frac{\varepsilon}{2M} s_n(x) \right\rfloor$ for all $x \in [0,1]$ and $n \geq 1$. These t_n are an increasing sequence of step functions. If $\{s_n\}_{n\geq 1}$ converges, then $\{t_n\}_{n\geq 1}$ converges as well and, hence, is eventually constant. If $\{s_n\}_{n\geq 1}$ diverges, then $t_{n+1}(x) - t_n(x) \geq 1$ for infinitely many values of n. Let

$$D_n = \{x \in [0,1] \mid t_{n+1}(x) - t_n(x) \geq 1\}. \tag{1.34}$$

Notice that $D \subseteq \bigcup_{n=1}^\infty D_n$. Now,

$$\int_0^1 (t_{n+1}(x) - t_n(x))\,dx \geq \int_0^1 \chi_{D_n}(x)(t_{n+1}(x) - t_n(x))\,dx \geq |D_n|\,. \tag{1.35}$$

This leads to

$$\begin{aligned}\sum_{n=1}^m |D_n| \leq \sum_{n=1}^m \int_0^1 (t_{n+1}(x) - t_n(x))\,dx &= \int_0^1 t_{m+1}(x)\,dx - \int_0^1 t_1(x)\,dx \\ &\leq \int_0^1 t_{m+1}(x)\,dx \leq \frac{\varepsilon}{2M}\int_0^1 s_{m+1}(x)\,dx < \varepsilon. \end{aligned}\tag{1.36}$$

Thus, $\sum_{n=1}^\infty |D_n| < \varepsilon$, and hence $|D| = 0$. The rest of the proof follows from the definition of upper functions. ■

August 10th.

The above is known as the monotone converge theorem for step functions. We can extend this to upper functions as follows.

Theorem 1.27. *Let $\{f_n\}_{n\geq 1}$ be an increasing sequence of upper functions such that $\lim_{n\to\infty}\int_0^1 f_n(x)\,dx$ exists. Then f_n converges pointwise almost everywhere to some upper function $f \in U[0,1]$ and $\int_0^1 f\,dx = \lim_{n\to\infty}\int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx$.*

Proof. Let $s_{n,k} \uparrow f_k$ almost everywhere, and define $t_n(x) = \max\{s_{n,1}(x), \ldots, s_{n,n}(x)\}$. Then t_n is increasing, and $t_n(x) \leq \max\{f_1(x), \ldots, f_n(x)\} = f_n(x)$ almost everywhere. Therefore, $\lim_0^1 t_n(x)\,dx \leq \int_0^1 f_n(x)\,dx \leq M$ for some M. Define $f(x) := \lim_{n\to\infty} t_n(x)$, wherever the limit exists. By the monotone convergence theorem for step function, f is defined almost everywhere, and $\int_0^1 f\,dx = \lim_{n\to\infty}\int_0^1 t_n(x)\,dx$. Also, $s_{n,k}(x) \leq t_n(x)$ for all $k \leq n$, so $f_k(x) = \lim_{n\to\infty} s_{n,k}(x) \leq \lim_{n\to\infty} t_n(x) = f(x)$ almost everywhere. Let $f_k(x) \to g(x)$ almost everywhere; we show that $f(x) = g(x)$ almost everywhere. Note that $g(x) \leq f(x)$ almost everywhere. Then $t_n(x) \leq f_n(x) \leq g(x)$ almost everywhere, so $f(x) \leq g(x)$ almost everywhere. Hence, $f(x) = g(x)$ almost everywhere. Hence, $f \in U[0,1]$. The rest of the proof follows from the monotone convergence theorem for step functions. ■

Again, we can extend this to Lebesgue integrable functions as follows.

Theorem 1.28 (The *monotone convergence theorem*)**.** *Let $\{f_n\}_{n\geq 1}$ be an increasing sequence of Lebesgue integrable functions such that $\lim_{n\to\infty}\int_0^1 f_n(x)\,dx$ exists. Then f_n converges pointwise almost everywhere to some Lebesgue integrable function $f \in L^1[0,1]$ and $\int_0^1 f\,dx = \lim_{n\to\infty}\int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx$.*
Alternatively, let $\{g_n\}_{n\geq 1}$ be a sequence of non-negative, almost everywhere, Lebesgue integrable func-

tions such that $\sum_{n=1}^{\infty}\int_0^1 g_n(x)\,dx$ *converges. Then the series* $\sum_{n=1}^{\infty} g_n(x)$ *converges almost everywhere to a Lebesgue integrable function* $g \in L^1[0,1]$*, and* $\int_0^1 g\,dx = \int_0^1 \sum_{n=1}^{\infty} g_n(x)\,dx = \sum_{n=1}^{\infty}\int_0^1 g_n(x)\,dx$.

We require the following theorem to prove the monotone convergence theorem for Lebesgue integrable functions.

Theorem 1.29. *Let* $f \in L^1[0,1]$ *and* $\varepsilon > 0$*. Then*

1. *there exist upper functions* $u, v \in U[0,1]$ *such that* $f = u - v$*, with* v *non-negative almost everywhere, and* $\int_0^1 v\,dx < \varepsilon$*, and*

2. *there exists a step function* $s \in \mathcal{S}[0,1]$ *and* $g \in L^1[0,1]$ *such that* $f = s + g$ *with* $\|g\|_1 < \varepsilon$.

Proof. 1. Let $f = u_1 - v_1$, with $u_1, v_2 \in U[0,1]$. Let $\{t_n\}_{n\geq 1} \subseteq \mathcal{S}[0,1]$ be such that $t_n \uparrow v_1$, and let $N \in \mathbb{N}$ be such that $0 \leq \int_0^1 (v_1 - t_N)(x)\,dx < \varepsilon$. Then $f = u_1 - t_N - (v_1 - t_N)$, with $u = u_1 - t_N$ and $v = v_1 - t_N$.

2. Let $f = u - v$, with $u, v \in U[0,1]$, and $v \geq 0$ almost everywere. Pick a step function $s \in \mathcal{S}[0,1]$ such that $0 \leq \int_0^1 (u-s)(x)\,dx < \frac{\varepsilon}{2}$. Then $f = u - v = s + (u - s) - v$, wth $g = u - s - v \in L^1[0,1]$. Then $\|g\|_1 = \int_0^1 |g(x)|\,dx \leq \int_0^1 (u-s)(x)\,dx + \int_0^1 v(x)\,dx < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

■

Proof of monotone convergence theorem. For $\varepsilon > 0$, let $g = u_\varepsilon - v_\varepsilon$, with $u_\varepsilon, v_\varepsilon \in U[0,1]$, $v_\varepsilon \geq 0$ almost everywhere, and $\int_0^1 v_\varepsilon\,dx < \varepsilon$, by the above theorem; for $n \in \mathbb{N}$, there exists a decomposition $g_n = u_n - v_n$, with $u_n, v_n \in U[0,1]$, $v_n \geq 0$ almost everywhere, and $\int_0^1 v_n\,dx < \frac{1}{2^n}$. Then $\sum_{n=1}^{\infty}\left(\int_0^1 v_n(x)\,dx\right)$ converges. Now, $u_n = g + v_n \geq 0$ almost everywhere; define $U_n(x) = \sum_{k=1}^{n} u_k(x)$, an increasing sequence of upper functions.Then

$$\int_0^1 U_n(x)\,dx = \sum_{k=1}^{n}\int_0^1 u_k(x)\,dx = \sum_{k=1}^{n}\left(\int_0^1 g(x)\,dx\right) + \sum_{k=1}^{n}\left(\int_0^1 v_k(x)\,dx\right). \tag{1.37}$$

However, both the sums on the right are finite; one by hypothesis, and one by the choice of the v_k's. Hence, $\lim_{n\to\infty}\int_0^1 U_n(x)\,dx$ exists. By the monotone convergence theorem for upper functions, U_n converges pointwise almost everywhere to U, an upper function on $[0,1]$, and $\int_0^1 U(x)\,dx = \lim_{n\to\infty}\int_0^1 U_n(x)\,dx$. Similarly, one can obtain V with $\int_0^1 V(x)\,dx = \lim_{n\to\infty}\int_0^1 V_n(x)\,dx$. Then, define $g = U - V$, which is Lebesgue integrable, and $\int_0^1 g\,dx = \sum_{n=1}^{\infty}\int_0^1 g_n(x)\,dx$. ■

This naturally leads to the next theorem.
August 17th.

Theorem 1.30 (The *dominated convergence theorem*)**.** *Let* $\{f_n\}_{n\geq 1}$ *be a sequence of Lebesgue integrable functions such that* $f_n \to f$ *almost everywhere, and there exists a Lebesgue integrable function* $g \geq 0$ *almost everywhere such that* $|f_n| \leq g$ *almost everywhere for all* $n \geq 1$*. Then* $f \in L^1[0,1]$ *and* $\int_0^1 f\,dx = \lim_{n\to\infty}\int_0^1 f_n(x)\,dx$.

Proof. Since f converges almost everywhere,

$$\lim_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x) = \inf_{n\geq 1}\sup_{k\geq n} f_k(x) = \lim_{n\to\infty}\sup_{k\geq n} f_k(x). \tag{1.38}$$

Now define $G_{n,i}(x) = \max\{f_i(x), \ldots, f_{i+n}(x)\}$ for $i, n \geq 1$. Then $|G_{n,i}| \leq g$ almost everywhere, and $\left|\int_0^1 G_{n,i}(x)\,dx\right| \leq \int_0^1 g(x)\,dx < \infty$. So for each fixed i, $\{G_{n,i}\}_{n\geq 1}$ is an increasing sequence of Lebesgue integrable functions. By the monotone convergence theorem, there exists $G_i \in L^1[0,1]$ such that $G_{n,i} \uparrow G_i$ and $\int_0^1 G_i(x)\,dx = \lim_{n\to\infty}\int_0^1 G_{n,i}(x)\,dx$. But note that $G_i(x)$ is simply $\sup_{k\geq i} f_k(x)$. Since $f_n \leq |f_n| \leq g$ almost everywhere, we have $G_i(x) \leq g(x)$ almost everywhere. Since the G_i's are decreasing, the sequence $\{g - G_i\}_{i\geq 1}$ is an increasing sequence of non-negative Lebesgue integrable functions, with $g - G_i \leq 2g$; by the monotone convergence theorem, there exists $h \in L^1[0,1]$ such that $g - G_i \uparrow h$ and $\int_0^1 h(x)\,dx = \lim_{i\to\infty} \int_0^1 (g - G_i)(x)\,dx$. But note that $h(x) = g(x) - \lim_{i\to\infty} G_i(x) = g(x) - \limsup_{n\to\infty} f_n(x) = g(x) - f(x)$ almost everywhere. Hence, $f = g - h$ is Lebesgue integrable, and

$$\int_0^1 f(x)\,dx = \int_0^1 g(x)\,dx - \int_0^1 h(x)\,dx = \int_0^1 g(x)\,dx - \lim_{i\to\infty} \int_0^1 (g - G_i)(x)\,dx = \lim_{i\to\infty} \int_0^1 G_i(x)\,dx. \tag{1.39}$$

■

Lemma 1.31 (*Fatou's lemma*)**.** *Let $\{f_n\}_{n\geq 1}$ be a sequence of non-negative Lebesgue integrable functions, and let $f(x) = \liminf_{n\to\infty} f_n(x)$. If $\liminf_{n\to\infty} \int_0^1 f_n(x)\,dx < \infty$, then f is Lebesgue integrable and $\int_0^1 f(x)\,dx \leq \liminf_{n\to\infty} \int_0^1 f_n(x)\,dx$.*

Proof. Let $g_n(x) = \inf_{k\geq n} f_k(x)$. Then $\{g_n\}_{n\geq 1}$ is an increasing sequence of non-negative Lebesgue integrable functions, with $g_n(x) \uparrow f(x)$ almost everywhere. By the monotone convergence theorem, there exists $g \in L^1[0,1]$ such that $g_n \uparrow g$ and $\int_0^1 g(x)\,dx = \lim_{n\to\infty} \int_0^1 g_n(x)\,dx$. But note that $g(x) = f(x)$ almost everywhere. Hence, f is Lebesgue integrable, and

$$\int_0^1 f(x)\,dx = \lim_{n\to\infty} \int_0^1 g_n(x)\,dx = \liminf_{n\to\infty} \int_0^1 f_n(x)\,dx. \tag{1.40}$$

■

Theorem 1.32. *$(L^1[0,1], \|\cdot\|_1)$ is a Banach space.*

Proof. Let $\{f_n\}_{n\geq 1}$ be a Cauchy sequence in $(L^1[0,1], \|\cdot\|_1)$. Then for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\|f_n - f_m\|_1 < \varepsilon$ for all $n, m \geq N$. In particular, we can choose $\varepsilon = 2^{-k}$ for $k \in \mathbb{N}$. Then there exists $N_k \in \mathbb{N}$ such that $\|f_n - f_m\|_1 < 2^{-k}$ for all $n, m \geq N_k$. We can assume that $N_k < N_{k+1}$ for all $k \in \mathbb{N}$. Then define $g_k(x) = \sum_{i=1}^k \left|f_{N_{i+1}}(x) - f_{N_i}(x)\right|$. Then $\{g_k\}_{k\geq 1}$ is an increasing sequence of non-negative Lebesgue integrable functions, with $\int_0^1 g_k(x)\,dx < \sum_{i=1}^k 2^{-i} < 1$. By the monotone convergence theorem, there exists $g \in L^1[0,1]$ such that $g_k \uparrow g$ and $\int_0^1 g(x)\,dx = \lim_{k\to\infty} \int_0^1 g_k(x)\,dx < 1$. Now define $h_n(x) = f_n(x) - f_{N_1}(x)$ for all $n \geq N_1$. Then $|h_n(x)| = |f_n(x) - f_{N_1}(x)| \leq g(x)$ almost everywhere. By the dominated convergence theorem, there exists $h \in L^1[0,1]$ such that $h_n \to h$ almost everywhere and $\int_0^1 h(x)\,dx = \lim_{n\to\infty} \int_0^1 h_n(x)\,dx$. Finally, define $f = h + f_{N_1}$. Clearly, $f_n \to f$ almost everywhere, and $\int_0^1 f(x)\,dx = \lim_{n\to\infty} \int_0^1 f_n(x)\,dx$. Hence, $f \in L^1[0,1]$, and $\{f_n\}_{n\geq 1}$ converges to f in $(L^1[0,1], \|\cdot\|_1)$. Therefore, $(L^1[0,1], \|\cdot\|_1)$ is a Banach space. ■

1.4 The Fourier Transform

Let $f \in L^1[0,1]$. Then, for $n \in \mathbb{Z}$, we define the n-th *Fourier coefficient* of f as

$$c_n := \int_0^1 f(x) e^{-2\pi i n x}\,dx. \tag{1.41}$$

This integral is well-defined since

$$|c_n| = \left|\int_0^1 f(x) e^{-2\pi i n x}\,dx\right| \leq \int_0^1 |f(x)|\,dx = \|f\|_1 < \infty \tag{1.42}$$

(note here that we have not yet proved the middle inequality for complex-valued functions; we shall assume it for now). Thus, for a function $f \in L^1[0,1]$, we define the function $\hat{f} : \mathbb{Z} \to \mathbb{C}$ as $\hat{f}(n) = c_n$ for all $n \in \mathbb{Z}$. This function is called the *Fourier transform* of f.

Given any two functions $f, g \in L^1[0,1] (\equiv L^1(S^1)$ via the periodic extension), we define the *convolution* of f and g as

$$(f * g)(x) := \int_0^1 f(x-y) g(y)\,dy \text{ for all } x \in [0,1]. \tag{1.43}$$

One may show that $f * g = g * f$.

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